Mahmoud Alyosify

AI Agent Systems Engineer

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SUMMARY

Engineer who builds **AI agents that finish multi-step work**: tool use, routing, self-checks and shared memory. At Injaz Tech, building a LangGraph agent behind a FastAPI service that routes questions, grades evidence and checks its own answers. Designed a multi-agent intelligence platform over a knowledge graph and an agent that turns PDFs into narrated slides. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30).

SKILLS

Agents: LangGraph, LangChain, tool and function calling, structured outputs, multi-agent systems, RAG, guardrails
LLMs: OpenAI and Anthropic APIs, Ollama, Hugging Face, prompt engineering, fine-tuning (LoRA, QLoRA)
Engineering: Python, FastAPI, Pydantic, pytest, Docker, GitHub Actions, AWS
Data: FAISS, ChromaDB, knowledge graphs, MongoDB, SQL
Evaluation: LLM-as-judge, retrieval metrics (top-1, MRR), automated evaluation gates

EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Built a **LangGraph** agent that routes each question, grades the evidence it finds, rewrites weak queries and checks its own answer.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.
- Built the **FastAPI** service: typed request and response models (Pydantic), streaming answers, session memory, health checks and rate limiting.
- Added prompt-injection filtering, output leak detection and rate limiting; **0 system-prompt leaks** in 8 attack runs.

Part-Time Machine Learning Instructor — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

AI in Cybersecurity Intern — Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) — Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.

Online Instructor — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

HORUS Sentinel — multi-agent threat-intelligence platform — 2026
Python, LangGraph, FastAPI, Neo4j, Ollama, Docker

- Designed a team of AI agents that keep collecting open-source information into a **knowledge graph**; a self-hosted LLM writes evidence-backed reports from it.
- Each agent owns one source; every finding is linked to earlier evidence in the shared graph, so reports trace each claim back to its source.

Automated Multimodal Agent — PDF to narrated slides — 2026
Python, LangGraph, Hugging Face, FAISS, Streamlit, text-to-speech

- An agent that turns a PDF into a narrated PowerPoint deck: it reads the document, plans and writes the slides, and records the narration.

Human vs. AI-Agent Pull Requests — research study
Python, GitHub data mining, statistics

- Built a reproducible pipeline that mines GitHub to compare review effort on pull requests written by AI agents and by people.

Horus-OSINT — fine-tuned LLM on AWS 2026
Python, Llama-3-8B, QLoRA, PySpark, AWS (EC2, S3, EMR), Terraform

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

MinimalLM — Master's thesis, mentored by Deloitte's AI team 2026
Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- MSc thesis: a training-free method that makes open LLMs generate **47.6% fewer tokens** and answer 19–68% faster, with no answer cut off.

Also: RL-Cloud-Autoscaler (PPO and DQN) · SimCLR-Vision-SSL (84.30% top-1, self-supervised) · AudioShield (deepfake audio detection).

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

CERTIFICATIONS

- AWS Certified Cloud Practitioner (Amazon Web Services)
- IBM AI Developer Professional Certificate (IBM, 2026)
- Generative AI: Prompt Engineering Basics (IBM, 2025)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI & Stanford, 2026)

AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University

LANGUAGES

Arabic (native) · English (professional working proficiency)